Week 1

Topic: Data Structure

1609. Even Odd Tree (Medium)

A binary tree is named **Even-Odd** if it meets the following conditions:

- The root of the binary tree is at level index 0, its children are at level index 1, their children are at level index 2, etc.
- For every **even-indexed** level, all nodes at the level have **odd** integer values in **strictly increasing** order (from left to right).
- For every **odd-indexed** level, all nodes at the level have **even** integer values in **strictly decreasing** order (from left to right).

Given the root of a binary tree, *return* true *if the binary tree is* ***Even-Odd****, otherwise return* false.

Example 1:

Input: root = [1,10,4,3,null,7,9,12,8,6,null,null,2]

Output: true

Explanation: The node values on each level are:

Level 0: [1]

Level 1: [10,4]

Level 2: [3,7,9]

Level 3: [12,8,6,2]

Since levels 0 and 2 are all odd and increasing and levels 1 and 3 are all even and decreasing, the tree is Even-Odd.

For detailed descriptions and constraints, please refer to Leetcode 1609.

25. Reverse Nodes in K-Group (Hard)

Given the head of a linked list, reverse the nodes of the list k at a time, and return *the modified list*.

k is a positive integer and is less than or equal to the length of the linked list. If the number of nodes is not a multiple of k then left-out nodes, in the end, should remain as it is.

You may not alter the values in the list's nodes, only nodes themselves may be changed.

Example 1:

Input: head = [1,2,3,4,5], k = 2

Output: [2,1,4,3,5]

Example 2:

Input: head = [1,2,3,4,5], k = 3

Output: [3,2,1,4,5]

For detailed descriptions and constraints, please refer to Leetcode 25.

3510. Minimum Pari Removal to Sory Array II (Hard)

Given an array nums, you can perform the following operation any number of times:

- Select the **adjacent** pair with the **minimum** sum in nums. If multiple such pairs exist, choose the leftmost one.
- Replace the pair with their sum.

Return the **minimum number of operations** needed to make the array **non-decreasing**.

An array is said to be **non-decreasing** if each element is greater than or equal to its previous element (if it exists).

Example 1:

Input: nums = [5,2,3,1]

Output: 2

Explanation:

- The pair (3,1) has the minimum sum of 4. After replacement, nums = [5,2,4].
- The pair (2,4) has the minimum sum of 6. After replacement, nums = [5,6].

The array nums became non-decreasing in two operations.

For detailed descriptions and constraints, please refer to Leetcode 3510.